



University Institute of Engineering
Department of Computer Science & Engineering

Experiment-7

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1. Aim of the practical: To study and analyze **functional dependencies and normalization of relational databases** by identifying candidate keys, prime and non-prime attributes, determining the highest normal form, and decomposing relations to reduce redundancy and improve database design.

2. Questions
(A)

Practice Set

1. Relation R(ABCDE) having dependencies as:

CE $\rightarrow$ D
D $\rightarrow$ B
C $\rightarrow$ A

Highest normal form? C.K set?

3. Relation R(ABCDEFGHI) having dependencies as:

AB $\rightarrow$ C
BD $\rightarrow$ EF
AD $\rightarrow$ GH
A $\rightarrow$ I

Highest normal form? C.K set?

5. Relation R(ABCDE) having dependencies as:

BC $\rightarrow$ ADE
D $\rightarrow$ B

Highest normal form? C.K set?

2. Relation R(ABCDEF) having dependencies as:

AB $\rightarrow$ C
DC $\rightarrow$ AE
E $\rightarrow$ F

Highest normal form? C.K set?

4. Relation R(ABCDE) having dependencies as:

AB $\rightarrow$ CD
D $\rightarrow$ A
BC $\rightarrow$ DE

Highest normal form? C.K set?

6. Relation R(ABC) having dependencies as:

A $\rightarrow$ B
B $\rightarrow$ C
C $\rightarrow$ A

Highest normal form? C.K set?



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Solution:

- 1. Candidate Key: CE | Highest Normal Form: 2NF**
- 2. Candidate Keys: ABD, BCD | Highest Normal Form: 1NF**
- 3. Candidate Key: ABD | Highest Normal Form: 1NF**
- 4. Candidate Keys: AB, BC, BD | Highest Normal Form: 3NF**
- 5. Candidate Keys: BC, CD | Highest Normal Form: 3NF**
- 6. Candidate Keys: A, B, C | Highest Normal Form: BCNF**

(B) The task is to design Student Performance Analysis System for a university, focusing on analyzing and evaluating student grades across courses. The system will track student names, course numbers, registration numbers, and grades. Functional dependencies include student name and course number determining grade, as well as registration number and course number determining grade. Additionally, student name determines registration number, and registration number determines student name. Identify the highest normal form for minimal redundancy by listing all the prime and non-prime attributes.

Solution:

Let the attributes be:

- S = Student Name
- C = Course Number
- R = Registration Number
- G = Grade

Functional Dependencies:

- $SC \rightarrow G$
- $RC \rightarrow G$
- $S \rightarrow R$
- $R \rightarrow S$



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Candidate Keys:

- {S, C}
- {R, C}

Prime Attributes: S, R, C

Non-Prime Attribute: G

Highest Normal Form: 3NF

(C) Consider a relation P having fields as F1, F2, F3, F4, F5 with the functional dependencies as follows:

- $F1 \rightarrow F3$
- $F2 \rightarrow F4$
- $(F1, F2) \rightarrow F5$

In terms of normalization, this table is in which normal form. Describe the prime and non-prime attributes for the relation P.

Solution:

Relation: P(F1, F2, F3, F4, F5)

Functional Dependencies:

- $F1 \rightarrow F3$
- $F2 \rightarrow F4$
- $(F1, F2) \rightarrow F5$

Candidate Key: {F1, F2}

Prime Attributes: F1, F2

Non-Prime Attributes: F3, F4, F5

Highest Normal Form: 1NF



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(D) DBA of a university information management system wants to remove redundancy by considering the following dependencies and relation $R(R_N, Name, Dob, Age)$ for the system. For better optimization of database, he needs to normalize the database by identifying the relevant CK keys and normal form.

- Dob $\rightarrow$ Age
- Name $\rightarrow$ Rno
- Age $\rightarrow$ Eligibility
- Rno $\rightarrow$ Name
- C_N $\rightarrow$ Cname
- C_N $\rightarrow$ Instructor
- (R_N, C_N) $\rightarrow$ Grade

Solution:

Functional Dependencies:

- Dob $\rightarrow$ Age
- Name $\rightarrow$ Rno
- Age $\rightarrow$ Eligibility
- Rno $\rightarrow$ Name
- C_N $\rightarrow$ Cname
- C_N $\rightarrow$ Instructor
- (R_N, C_N) $\rightarrow$ Grade

Candidate Keys:

- {R_N, C_N, Dob}
- {Name, C_N, Dob}

Prime Attributes: R_N, Name, C_N, Dob

Non-Prime Attributes: Age, Eligibility, Cname, Instructor, Grade

Highest Normal Form: 1NF



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3. Learning Outcomes:

- Learned to identify functional dependencies and candidate keys.
- Gained understanding of prime and non-prime attributes.
- Learned to determine the highest normal form of a relation.
- Developed practical skills in identifying partial and transitive dependencies.
- Learned how normalization helps reduce redundancy and improve database design.